NORTHSTAR

Physical AI | Autonomous Robotics | Industrial Intelligence

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01. HERO SECTION

Intelligence That Moves Beyond the Screen.

Physical AI for the Real World.

We bring intelligence into the physical world through autonomous drones, quadruped robots, humanoid robotics, and connected industrial systems.

Powered by our agentic AI platform, Northstar transforms how enterprises observe, inspect, verify, and understand their physical environments — connecting autonomous machines with intelligent decision-making to make industrial operations safer, smarter, and more predictable.

[Explore Our Platform] [Talk to Our Team]

Autonomous Inspection. Continuous Intelligence. Verified Reality.

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02. INTRODUCTION — THE NORTHSTAR VISION

The Physical World Is Ready for Intelligence.

For decades, enterprise technology has focused on digitizing information. The next frontier is enabling intelligence to interact with the physical world.

Northstar is building the intelligence layer between AI and the real world.

By combining embodied AI, autonomous robotics, computer vision, spatial intelligence, and IoT, we enable machines to capture real-world conditions, understand complex environments, and execute coordinated inspection and verification workflows.

From construction sites and industrial facilities to energy infrastructure and critical assets, our vision is to turn physical operations into continuously observable, verifiable, and increasingly autonomous systems.

We don't just capture the physical world. We enable AI to understand it.

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03. OUR TECHNOLOGY - AUTONOMOUS SYSTEMS

Intelligence in Every Dimension.

A connected ecosystem of intelligent machines.

Northstar brings aerial robotics, ground robotics, humanoid systems, and distributed sensors together through one intelligent orchestration platform.

Autonomous Drones

Intelligence from Above.

Capture, map, monitor, and inspect complex environments from the air.

Our drone-enabled workflows support aerial surveying, photogrammetry, LiDAR scanning, thermal imaging, infrastructure inspection, volumetric analysis, and construction progress monitoring.

From large construction developments to remote industrial infrastructure, aerial intelligence helps teams understand changing site conditions without relying exclusively on manual inspections.

Capabilities: Aerial Reality Capture | *Mapping & 3D Reconstruction* | *Progress Monitoring* | *Thermal Inspection* | *Asset Verification*

Quadruped Robotics

Intelligence on the Ground.

Four-legged robotic platforms extend inspection capabilities into complex industrial and construction environments.

Equipped with visual cameras, depth sensors, thermal imaging, and spatial navigation capabilities, quadruped robots can support repeatable inspection missions, indoor reality capture, equipment monitoring, and remote site observation.

They enable consistent data collection across large facilities, plant rooms, corridors, and other challenging operational environments.

Capabilities: Autonomous Patrols | *Ground Reality Capture* | *Thermal Monitoring* | *Equipment Inspection* | *Remote Observation*

Humanoid Robotics

Intelligence That Works Alongside People.

Humanoid robotics represents the next evolution of embodied intelligence, bringing AI-assisted perception and manipulation into human-designed environments.

Northstar is developing workflows for humanoid systems to support supervised visual inspections, equipment interaction, inspection checklist execution, material identification, and physical task assistance.

Our approach combines visual reasoning, task planning, robotic control, and human oversight to progressively extend the role of autonomous systems in industrial operations.

Focus Areas: Embodied AI | *Visual Inspection* | *Human-Robot Collaboration* | *Task Planning* | *Physical Workflow Assistance*

Industrial IoT & Edge Intelligence

Intelligence from Every Connected Asset.

Not every inspection requires a moving robot.

Northstar connects industrial sensors, cameras, edge devices, and operational systems to create a continuous flow of physical-world intelligence.

By bringing together machine telemetry, environmental measurements, asset conditions, and robotic observations, we establish a unified operational view of facilities and infrastructure.

Capabilities: Sensor Integration | *Edge AI* | *Condition Monitoring* | *Anomaly Detection* | *Equipment Telemetry*

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04. THE NORTHSTAR PLATFORM

One Intelligence Platform. Every Physical System.

Meet Northstar Physical AI Platform.

An agentic intelligence and robotics orchestration platform designed to connect physical-world observations with autonomous workflows, intelligent analysis, and operational decision-making.

Northstar serves as the coordination layer between machines, sensors, enterprise information, and the people responsible for operating physical assets.

Perceive. Reason. Orchestrate. Verify.

Autonomous Mission Orchestration

Plan, assign, and coordinate inspection missions across supported drones, ground robots, and connected systems. Manage recurring capture activities and mission execution through centralized workflows.

Multimodal Perception & Spatial Intelligence

Combine imagery, video, LiDAR, point clouds, thermal data, and sensor telemetry to build contextual understanding of sites, structures, equipment, and physical environments.

Agentic AI & Intelligent Reasoning

AI agents interpret observations against inspection requirements, drawings, asset records, operating procedures, and historical evidence to identify anomalies, prioritize issues, and recommend follow-up actions.

AI-Powered QA & Verification

Compare observed site conditions with defined specifications, BIM models, inspection checklists, and quality requirements. Generate evidence-linked findings, identify possible deviations, and support verification workflows.

Digital Twin & Reality Intelligence

Connect captured site conditions with spatial models and digital asset records to maintain a contextual view of the physical environment and its changes over time.

Enterprise Integration & Human Oversight

Connect intelligence with construction, asset management, maintenance, and enterprise systems. Keep humans in control of safety-critical decisions, corrective actions, and formal approvals.

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05. HOW NORTHSTAR WORKS

From Physical Observation to Intelligent Action.

A continuous intelligence loop connecting the real and digital worlds.

01 — Plan

Define inspection objectives, target assets, operating constraints, schedules, and verification requirements.

02 — Deploy

Assign approved missions to drones, robotic systems, or connected sensors, with the appropriate autonomy and human supervision.

03 — Capture

Collect spatially referenced visual, thermal, geometric, and operational data from the physical environment.

04 — Understand

Use multimodal AI, computer vision, and contextual reasoning to identify conditions, anomalies, changes, and potential deviations.

05 — Verify

Compare findings against drawings, BIM models, specifications, previous inspections, and defined acceptance criteria.

06 — Act & Learn

Create actionable findings, initiate review workflows, generate reports, and use verified outcomes to improve future inspection processes.

From data capture to evidence-based decisions — connected by intelligence.

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06. FEATURED SOLUTION — CONSTRUCTION INTELLIGENCE

Every Site. Every Stage. Verified.

AI-powered construction quality assurance and progress intelligence.

Construction environments are dynamic, complex, and constantly changing. Traditional inspections are periodic, heavily manual, and often fragmented across teams and systems.

Northstar combines aerial reality capture, ground robotics, computer vision, and agentic AI to support more consistent, evidence-driven construction monitoring and quality assurance.

Our goal is to provide contractors, developers, engineers, and project owners with a clearer understanding of what has been built, how work is progressing, and where intervention may be required.

Construction Progress Intelligence

Monitor site changes over time using aerial and ground-level observations. Compare captured progress with project schedules, work packages, and planned construction activities.

Build-versus-Design Verification

Use reality capture and spatial analysis to compare observed construction conditions with BIM models, approved drawings, and design specifications, identifying potential discrepancies for engineering review.

Automated QA/QC Inspections

Support inspection checklists, workmanship observations, installation verification, defect identification, and evidence-based quality reporting throughout the construction lifecycle.

Safety Observation & Risk Intelligence

Identify visible safety concerns and potential site hazards through AI-assisted observations, helping safety teams prioritize investigations and corrective measures.

Quantity & Work-in-Place Verification

Support earthwork volume measurements, installation progress assessment, and verification of work-in-place quantities where capture accuracy and measurement methods meet project requirements.

Digital As-Built Documentation

Build a chronological, spatially referenced record of site conditions, supporting handover documentation, issue resolution, project coordination, and future asset management.

Inspection Evidence & Traceability

Maintain georeferenced, time-stamped observations with associated images, inspection findings, responsible parties, verification status, and supporting documentation.

See what is happening. Understand what has changed. Verify what has been delivered.

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07. BEYOND CONSTRUCTION — INDUSTRIAL APPLICATIONS

Built for Environments Where Precision Matters.

Industrial Assets & Manufacturing

Support equipment condition monitoring, visual quality inspections, thermal anomaly detection, production-facility walkthroughs, and asset documentation.

Energy & Utilities

Inspect selected energy infrastructure, solar installations, substations, and utility assets through aerial imaging, thermal analysis, connected sensors, and robotic observation.

Oil, Gas & Process Facilities

Support remote visual inspection, equipment condition assessment, thermal observations, and routine monitoring in suitable industrial operating environments.

Infrastructure & Civil Engineering

Monitor roads, bridges, rail corridors, tunnels, and large-scale civil assets through reality capture, structural-condition observations, and change detection.

Warehouses & Logistics Facilities

Enable inventory-area observation, facility inspection, equipment monitoring, and spatial documentation through autonomous mobile platforms and connected intelligence.

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08. WHY NORTHSTAR

Beyond Automation. Towards Physical Intelligence.

One Platform, Multiple Robotic Systems

Bring diverse robots, drones, sensors, and inspection data into a coordinated intelligence environment.

Designed for Real-World Complexity

Combine spatial perception, contextual understanding, and adaptive workflows for dynamic physical environments.

Evidence Before Assumptions

Ground inspection findings in captured observations, relevant specifications, timestamps, and traceable verification records.

Autonomy with Accountability

Balance automated missions and AI reasoning with safety controls, human supervision, and defined approval processes.

From Individual Assets to Enterprise Scale

Apply consistent inspection and intelligence workflows across assets, facilities, projects, and distributed operations.

Intelligence That Connects to Enterprise Workflows

Move beyond dashboards by linking physical observations to quality, maintenance, asset, and operational management processes.

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09. ABOUT NORTHSTAR

Engineering the Future of Physical AI.

Northstar is a physical AI and autonomous systems company focused on connecting machine intelligence with real-world operations.

We are bringing together agentic AI, robotics, computer vision, spatial computing, and industrial connectivity to reimagine how enterprises inspect, verify, and manage physical environments.

Our vision is a future where intelligent systems continuously understand the world around them, collaborate with human teams, and help enterprises make better decisions through reliable physical-world evidence.

Our Mission

To make physical environments more observable, intelligent, and verifiable through autonomous systems and AI.

Our Vision

A world where intelligence moves seamlessly between digital systems and physical reality.

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10. FINAL CTA

The Future of Intelligence Is Physical.

Bring autonomous intelligence into your operations.

From construction quality assurance to industrial inspection and intelligent asset monitoring, Northstar is building the next generation of physical AI solutions.

[Explore Northstar] [Partner With Us]

Let's build a world where machines don't just operate — they understand.

FOOTER

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Intelligence Beyond the Digital World.